CHILD DEVELOPMENT

EMPIRICAL ARTICLE

Maternal Sensitivity Predicts Child Attachment in a Non-Western Context: A 9-Year Longitudinal Study of Chinese Families

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ABSTRACT

Despite the long-standing debate over the assumed universality of maternal sensitivity predicting attachment security (i.e., sensitivity hypothesis), few long-term longitudinal investigations on attachment have been conducted outside the Western context. We leveraged data from a prospective 9-year longitudinal study of middle-class families ($N = 356$; female = 48.9%) in China to examine if early maternal sensitivity predicts attachment representations in middle childhood. Maternal sensitivity was assessed from lab-based observed interactions at 14 and 24 months. At 10 years old, children completed the Chinese version of the Attachment Script Assessment. Maternal sensitivity positively predicted the child's attachment representations at age 10 years ($\beta = 0.20$, $p < 0.01$). These results supported the view that maternal sensitivity is prospectively related to secure attachment across cultures.

1 | Introduction

Bowlby-Ainsworth's attachment theory conceptualizes the parent–child bond as a secure base relationship (e.g., Bowlby 1969/1982). The theory highlights that a caregiver's consistent, appropriate, and sensitive responses to the child's needs for support during exploration and in response to distress foster the expectation of successful support in the child (i.e., the sensitivity hypothesis; Ainsworth 1979; Ainsworth et al. 1978/2015; Bowlby 1973). Frequent experiences of sensitive support thereby facilitate secure attachment development and shape the organization of secure base behavior (Ainsworth et al. 1978/2015). Forming a secure attachment to primary caregivers is seen as a product of evolutionary adaptation and therefore culturally universal (Bowlby 1988; van Ijzendoorn and Sagi-Schwartz 2008). Nevertheless, attachment theory has encountered criticism for its limited applicability and its disregard for non-Western caregiving practices and philosophies (e.g., Gaskins et al. 2017; see Waters & Eagle, submitted for full discussion of these criticisms). The debate surrounding the cross-cultural generalizability of attachment theory is long-standing (see: Harkness 2015; Keller 2018, 2019; Mesman et al. 2016;

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Rothbaum et al. 2000), yet the availability of long-term longitudinal studies in non-Western contexts to evaluate its validity has remained limited. In this study, we present an evaluation of attachment theory's cross-cultural generalizability utilizing longitudinal data from an urban Chinese context.

Individual differences in a child's attachment security are said to arise from the differences in the quality and consistency of the caregiving they receive (i.e., the sensitivity hypothesis; Ainsworth 1979; Ainsworth et al. 1978/2015; Bowlby 1973, 1988). Children who receive prompt and consistent responses from caregivers to their bids for support during distress and exploration develop a secure attachment representation, characterized by positive expectations regarding caregiver availability, responsiveness, and sensitivity (Bowlby 1980). In contrast, children with experiences of inconsistent, insensitive, or even harsh and rejecting care form an insecure attachment which diminishes their confidence in their caregivers as a secure base for exploration and as a source of comfort/support (e.g., Ainsworth et al. 1978/2015; Cyr et al. 2010). These early caregiving experiences are internalized as a mental representation of attachment, referred to as an internal working model (IWM; Bowlby 1988). The IWM of attachment is thought to consist of various constructs related to our representations of self and other in the context of attachment relationships including a cognitive script summarizing routine attachment support (Bowlby 1969/1982). This cognitive script, termed the secure base script, is a tempero-casual set of expectations generalized from repeated secure base interactions where the child will signal for support when in distress and the attachment figure will help to effectively resolve the child's distress followed by a return to normal (Waters and Waters 2006; Waters et al. 2021). The secure base script, and IWM more broadly, is thought to reflect a set of generic expectations of attachment relationships rather than any specific relationship (Boldt et al. 2015; Fivush 2006; Waters, Fraley, et al. 2015). Prior work indicates that the secure base script, like any cognitive script, is learned through recurring experience, is stable over time, and influences behavior (see Waters and Roisman 2019 for a review). Furthermore, research implicates these early relationship expectations in determining children's cognitive, emotional, and behavioral development (e.g., Deneault et al. 2023; Groh et al. 2017; Posada et al. 2019).

Importantly, Ainsworth and Bowlby posited secure attachment formation as a universal fundamental for individual human development on the basis that the development of attachment is thought to stem from an evolutionary and biologically adaptive need to survive (Ainsworth 1979; Bowlby 1969/1982). Forming secure attachments to a primary caregiver enhances the infant's chances of survival by seeking proximity to their caregiver who provides a safe haven for the child when threatened; in turn, the caregiver may also provide a secure base for the infant's exploration once safe to do so (Bowlby 1969/1982, 1973, 1980). This posited universality of forming secure attachment bonds is upheld by four central hypotheses (van Ijzendoorn and Sagi-Schwartz 2008), namely: (1) that all infants will become attached to one or more attachment figures (i.e., the universality hypothesis); (2) that secure attachment is considered normative under most circumstances (i.e., the normativity hypothesis); (3) sensitive-responsiveness or parental sensitivity is a crucial factor in establishing secure attachment (i.e., the sensitivity hypothesis); and (4) that infants who are securely attached will exhibit higher socio-cognitive development as compared to their insecurely attached counterparts (i.e., the competence hypothesis). These tenets highlight the development of attachment as an evolutionarily adaptive behavior system (see Simpson and Jaeger 2022 for review). Within the central tenets of attachment's universality, the sensitivity hypothesis in particular has been the subject of much interest within academic discourse; in particular, researchers have sought to substantiate as well as disprove the claim.

Attachment researchers have conducted extensive cross-cultural work to address the universality of the sensitivity hypothesis (see Fourment et al. 2022; Mesman et al. 2016 for a review). Studies largely support the notion that maternal sensitivity (and maternal attachment security) contributes to the formation of a secure attachment in the next generation across cultures (De Wolff and van IJzendoorn 2006; van Ijzendoorn and Sagi-Schwartz 2008). Recent meta-analytic findings offer support for the link between parental sensitivity and child attachment security, with an overall effect size of $r = 0.25$ (Madigan et al. 2024). Despite extensive cross-cultural research, there are some notable limitations, including a predominant focus on cultures within the Western world, leading to an underrepresentation of non-Western cultures as well as an overemphasis on cross-sectional studies for examining attachment transfer, rather than long-term longitudinal research (e.g., Carlson and Harwood 2003; Ding et al. 2012; Posada et al. 2016), which would provide stronger evidence when examining the caregiving elements that influence secure attachment. Of the longitudinal studies conducted on attachment development, the overwhelming majority have been situated in Western contexts, predominantly in the United Kingdom (e.g., Booth-LaForce and Roisman 2014; Sroufe et al. 2005; Waters et al. 2000) and Western Europe (e.g., Grossmann et al. 2005; Schoenmaker et al. 2015; Steele and Steele 2005). This significantly limits our ability to evaluate the sensitivity hypothesis outside Western contexts.

1.1 | A Brief History of the Critiques of the Sensitivity Hypothesis

The sensitivity hypothesis has drawn several distinct criticisms from a cross-cultural cultural standpoint: (1) the role of sensitive maternal care, in particular maternal warmth, is overstated in forming secure attachments and (2) nurturing autonomy and independence through support for exploration reflects uniquely Western individualistic concerns. With these assertions, critics argue that attachment theory seemingly neglects differences in value systems, parenting beliefs, and cultural parenting practices in non-Western contexts, challenging the putative universality of the sensitivity hypothesis (e.g., Dawson 2018; Harkness 2015). Following this, the ethics of utilizing attachment theory to inform global parenting interventions or those targeting non-Western families have been severely scrutinized (e.g., Morelli 2015; Morelli et al. 2018; Yeo 2003).

Sensitive, emotionally warm, and responsive parenting is widely regarded as a key component in secure attachment development (Ainsworth et al. 1978/2015; Waters et al. 1979). Scholars indicate that emotional warmth, along with mutual responsivity, are distinct parenting concepts that influence relational quality between parent and child (Laible et al. 2015). Although not originally outlined in Ainsworth's sensitivity observational scale (Ainsworth et al. 1978/2015), warmth, or positive affect, has become a modern inclusion in commonly used and widely validated observational caregiving sensitivity assessments (see NICHD Early Child Care Research Network, P.I.C.B 2001; Fraley et al. 2013; Roisman et al. 2009). Maternal warmth is thought to encourage the child to continue to seek out secure base support and allow the child to feel a sense of trust and safety in the secure base figure (Ainsworth et al. 1978/2015; Mesman and Emmen 2013). However, some anthropologists and psychologists contend that this emphasis on warm and positive maternal emotional interactions is not aligned with the caregiving practices prevalent in various non-Western, economically diverse, and rural cultures (e.g., Harkness 2015; Keller et al. 2018; Morelli 2015; Richman et al. 1992).

Critics of the sensitivity hypothesis also take issue with the focus on autonomy support and the child as the central figure in the attachment relationship who "takes the lead" (Keller 2018). Ainsworth described an equilibrium at the center of attachment behavior. The child contends with their need for proximity to their attachment figure and their exploratory behavioral system (Ainsworth 1982; Ainsworth et al. 1978/2015). The link between caregiving support and autonomous exploration in the child is believed to be an evolved universal process critical for development and survival (e.g., establishing social connections and managing risk; Bowlby 1973; Simpson and Jaeger 2022). Counterarguments reference various anthropological studies that highlight different non-Western caregiving practices, wherein parents prioritize the cultivation of a communal agent over an independent one (Rothbaum et al. 2000; Quinn and Mageo 2013); stating that these practices are incongruent with Bowlby and Ainsworth's descriptions of secure base behavior and the role of caregiver sensitivity in the context of supporting autonomous exploration.

However, these critiques of the sensitivity hypothesis rely on *group-level* observations about caregiving practices and beliefs across entire cultures, rather than accounting for *individual* differences between caregiver–child dyads (Dawson 2018; Rothbaum et al. 2000). Certainly, non-Western cultures vary at the group level on the extent to which they emphasize maternal warmth or autonomy-supporting care (e.g., Keller et al. 2007; Liu et al. 2005; Wu et al. 2002). Yet, these arguments have two noteworthy shortcomings: (1) they conflate the *goals* and *means* of caregiving. Attachment theory leaves room for cultural variation in the socialization goals of caregivers, including variation in the level of autonomy expected. The consistency and attunement of the caregiver's responses to infant signals is thought to be the source of the child's attachment expectations, not their caregiver's socialization goals (see Waters and Eagle submitted for further discussion). From the attachment perspective, maternal sensitivity represents the means by which a caregiver encourages and supports the child to engage with and learn about the world and guides a child toward the socialization goals of the community and culture and (2) they describe mean level differences across cultures as a problem for a theory of individual differences. Even if sensitivity were markedly lower at the mean level in a particular cultural context, this does not nullify the sensitivity hypothesis unless individual differences in caregiving sensitivity no longer predict individual differences in infant or child secure base representations and behavior in that context.

Although many of attachment theory's critics call for an indigenous and emic consideration of sensitive parenting, a majority of their findings are based on anthropological, naturalistic observation, or unstructured discussions with relatively small sample sizes and limited training in assessments of maternal sensitivity or infant/child secure base behavior and security (e.g., Chapin 2013; Yeo 2003). Few studies have empirically assessed their claims in refuting the sensitivity hypothesis and its universality utilizing well studied assessment tools in adequately powered, longitudinal, non-Western datasets. The scarcity of non-Western attachment data could be attributed to a lack of culturally and linguistically adaptable empirical forms of assessment, particularly in measuring attachment representations. Therefore, there is a significant research gap not only in empirically examining the influence of caregiver sensitivity as a developmental antecedent of attachment security within distinctly non-Western contexts, characterized by unique parenting styles and philosophies, but also in developing or culturally tailoring research instruments to investigate this relation.

1.2 | China—A Distinct Parenting and Developmental Context

According to the critiques, the sensitivity hypothesis would likely not be supported in the context of China because, on the group level, China differs from most Western countries in historical, political, and cultural orientations (Hofstede 2011; Luo et al. 2013; Yang 1995). Prior research suggests that the expression of warmth is more restricted in Chinese families as it is seen as undermining parental authority (e.g., Chen et al. 1998; Wu and Chao 2005). Moreover, when compared to their Western counterparts, Chinese parents tend to more frequently employ strategies related to psychological control, including guilt induction, authority assertion, and love withdrawal, which are seen as inconsistent with the sensitivity construct (e.g., Chao 1995; Cheung and Pomerantz 2011; Luo et al. 2013; Ng et al. 2013). It is important to note that emerging research suggests that the differences between Chinese and Western parents are not as distinct as perhaps initially characterized (Cheah et al. 2013; Chen et al. 1998; Way et al. 2013). However, China is still widely considered a child development context that differs substantially from the West, especially along attachment-relevant constructs (e.g., warmth, autonomy support, psychological control).

The construct of parental sensitivity has been cross-culturally assessed and applied in several studies across East Asian contexts (e.g., Singapore: Cheung 2021; Cheung and Elliott 2016 and Japan: Miyake et al. 1985; Mizuta et al. 1996; Vereijken et al. 1997). Previous work on both Chinese and Korean families indicated that mothers whose infants were classified as secure during the Strange Situation Procedure were more sensitive toward their children than the mothers of infants classified as insecure (Ding et al. 2012; Jin et al. 2012). These findings indicate a cross-cultural association between parental sensitivity and attachment development. However, to date, much of the existing research has employed cross-sectional designs with moderate sample sizes, limiting the ability to assess the predictive value of maternal sensitivity on secure attachment development. For example, in a short-term longitudinal study conducted by Liang et al. (2021) examining the effects of the Chinese grandmother–mother–infant caregiving network on infant psychosocial adjustment, maternal sensitivity and mother–infant attachment were assessed at 12 months and were found to be significantly associated. A few studies also support attachment theory more broadly outside the Western context by indicating that higher maternal sensitivity is associated with fewer behavioral problems in infants/children rather than assessing security as the outcome (Li et al. 2022; Xing et al. 2016; Zhang and Wang 2022). In contrast, some studies with Chinese parents living in other countries find that maternal sensitivity is associated with greater child behavior problems and lower likability by peers (Chan et al. 2010; Cheung and Elliott 2016), raising questions about the meaning of parental sensitivity in Chinese culture. It is imperative to note that the mixed results in the literature primarily come from cross-sectional designs with limited sample sizes (but see Zhang and Wang 2022), leaving the role of maternal sensitivity in East Asian families (and more specifically, Chinese families) in need of further study. To our knowledge, there have been no long-term, longitudinal studies of attachment and maternal sensitivity in a non-Western context. In order to enhance our understanding of sensitive caregiving outside Western contexts, cross-cultural attachment research would be greatly enhanced by longitudinal data and sufficiently large sample sizes to more precisely estimate effect sizes and improve causal inference relative to cross-sectional designs.

1.3 | The Present Study

The current study aims to provide longitudinal evidence in the academic discourse surrounding the universality of the sensitivity hypothesis. We seek to do this by examining maternal sensitivity during infancy/toddlerhood as a developmental predictor of middle childhood secure attachment representations in an urban Chinese sample. To do so, we first adapted the middle childhood version of the Attachment Script Assessment (ASA; Waters, Bosmans, et al. 2015) for the Chinese cultural context. The ASA assesses an individual's knowledge and access to the secure base script, a core component of a secure IWM (Waters and Roisman 2019; Waters and Waters 2006). The ASA represents a culturally versatile methodological tool as it relies on simple outlines about widely experienced caregiver–child and relationship scenarios (e.g., getting physically hurt) to determine secure base script knowledge. As such, it can be adapted for different cultural contexts either through direct translation or by modifying the narratives to reflect culturally normative contexts for secure base use and support. The ASA has been successfully adapted for Dutch (e.g., Verhees et al. 2019; Waters, Bosmans, et al. 2015), Portuguese (Monteiro et al. 2008; Verissimo and Salvaterra 2006), and Japanese contexts (Umemura et al. 2018).

Furthermore, within the Chinese context, secure base script knowledge partially mediated the relation between early caregiving and social adjustment in middle childhood (Yang et al. 2024). We constructed four culturally appropriate scenarios for urban Chinese children (i.e., two mother–child and two father–child items) along with one neutral practice story to limit hypothesis guessing (Waters and Waters 2021). To gauge the success of our efforts in adapting the ASA to the Chinese context, we assessed the internal consistency of the ASA items and tested for replication of Waters, Bosmans, et al. (2015) to demonstrate the relative fit of our adapted ASA to a generalized model. We hypothesized that mother and father items would generalize to a single underlying factor, as has been observed in previous research.

We then leveraged data collected 9 years earlier as part of the Nanjing Child Development Project (NCDP; Way et al. 2013; Yang et al. 2024; Zhang et al. 2019) to conduct a longitudinal examination of the sensitivity hypothesis within the Chinese context. Our specific focus was to investigate whether the early caregiving quality in Chinese mothers contributes to the development of secure base script knowledge in middle childhood. The NCDP is a prospective longitudinal study of social and emotional development that began in 2006 in the city of Nanjing, China. A sample of normative-risk mothers and their babies (all singletons) were recruited for the study and participated in several lab-based assessments including two observations of mother–child interactions when the target child was 14 and 24 months old. At age 10 years ($N = 256$), the target child returned to the lab and completed a battery of assessments including the Chinese middle childhood version of the ASA. Specifically, we hypothesized that a composite of maternal sensitivity coded from the 14 month and 24-month mother–child observations would positively predict ASA scores at age 10 years. Previous research has found maternal education to be significantly associated with maternal sensitivity (e.g., Garai et al. 2009; Tamis-Lemonda et al. 2009). Moreover, Raby et al. (2015) found that child's biological sex and maternal education partially account for the predictive significance of early maternal sensitivity on later adjustment and much of the prior research that used secure base script knowledge as an outcome included child gender/biological sex and/or maternal education as covariates (e.g., Vaughn et al. 2016). As such, we take maternal education and child's biological sex into account as covariates in our analysis to allow for comparison of our results with the extant literature.

2 | Methods

2.1 | Participants

Data were drawn from an ongoing longitudinal mixed-method study on parenting and child development in the changing social context in Nanjing, China. Nanjing is the capital city of Jiangsu province and has experienced rapid economic development and modernization in the past few decades. During the time of our data collection from 2007 to 2016, Nanjing's GDP grew from 320.7 billion RMB (approx. 42.76 billion USD) to 1.17 trillion RMB (approx. 180.2 billion USD) in 2016 (Nanjing Municipal Bureau Statistics 2008a, 2017a). The population in Nanjing grew from 7.41 million in 2007 to 9.14 million in 2016 (Nanjing Municipal Bureau Statistics 2008b, 2017b).

During 2006–2007, 438 first-time mothers ($M_{age} = 28.96$, $SD = 3.02$) were recruited from a municipal hospital in the central area of Nanjing. The sample is predominantly middle-upper class, where mothers (78.08%) and fathers (78.54%) reported attaining a college degree or higher education level. By the time the child was 6 months old, 80.7% of mothers had resumed working. Ethnically, the families are predominantly Han (95.09%), and 4.90% are from minority ethnic groups.

While altogether there have been seven waves of data collection that took place from 2007 to 2021, for the purpose of the present study, we focused on mother–child interactions collected at 14- and 24-month lab visits and secure base script knowledge assessed on the target child at the age 10-year follow-up. At 14 months old, participants were invited to the lab. Due to the resources and funding available, half of the participants were randomly selected to participate in video-recorded mother–child interactions. A total of 169 dyads completed this follow-up. At the 24-month visit all families were invited to participate in video-recoreded mother-child interactions with 282 dyads completing this assessment. At age 10, 251 children participated in an in-lab data collection procedure which included the ASA.

To preserve the maximum sample size, we included participants who completed at least one of the three waves' tasks at 14 months, 24 months, or 10 years. Therefore, the final sample size for this study consists of 356 mother–child dyads, among which 48.88% of the children are female. In the final sample size, 72.41% of mothers hold an undergraduate degree, 9.20% hold a postgraduate degree, and 18.39% hold a secondary school degree. Eleven families (4.38%) reported divorce in 10 years old follow-up. Monetary compensation for each wave varied based on the tasks and time commitment required in each session. No significant differences were found in child's sex and maternal education level between participants in our final sample and those who were initially recruited. All in-lab data collection took place in a child development laboratory at a university in Nanjing. IRB approval was obtained from this university. Parental consent forms were collected at each wave, and at the age 10-year follow-up, children's assent was also collected.

2.2 | Measures

2.2.1 | Maternal Sensitivity

We assessed maternal sensitivity at 14 and 24 months through lab-based observations of mother–child interactions, using scales developed for the NICHD Study of Early Child Care and Youth Development (NICHD Early Child Care Research Network 1999, 2003) based upon the Qualitative Scales of the Observational Ratings of the Caregiving Environment (ORCE; Owen 1992). At the 14-month observation, maternal sensitivity was coded from a 10-min free-play session. At the 24-month observation, maternal sensitivity was coded from an 11-min mother–child interaction consisting of a 5-min free-play session, a 4-min book-reading task, and a 2-min bead threading task. In all sessions, mothers were instructed to play or read with their children in whatever ways they liked.

The raters coded three dimensions of maternal behavior using five-point scales: sensitivity/responsiveness to non-distress, intrusiveness, and positive regard. Sensitivity/responsiveness to non-distress assesses the extent to which the mother is attuned to her child's cues and allows this awareness to guide their interaction. Typical behaviors scored high for the sensitivity/responsiveness to non-distress scale include contingently responding to the child's behaviors and expressions, providing stimulation that is appropriate to the situation based on the child's cues, and following and building on the child's focus of attention. Intrusiveness assesses how much mothers direct the interaction or disrupt the child's behavior without accommodating the child's needs. Behaviors such as imposing an agenda on the child, neglecting signals from the child that a different activity or a different pace of interaction is needed, or verbally/physically overstimulating the child are coded as high on intrusiveness. Mothers will also score high on intrusiveness if they overwhelm their children with a rapid succession of toys or suggestions and leave no space for children to react to one stimulus before another is presented. The positive regard domain refers to the level of positive feelings expressed toward the child during interaction. A warm tone of speaking, expressions of physical affection, enjoyment and enthusiasm toward the child, and praise of the child are all typical indicators of positive regard for the child.

All videos were double-coded independently by two trained coders, who are both native to Nanjing. Any disagreements larger than one point were resolved through discussion until a consensus was reached. The intraclass correlations between the two coders ranged from 0.79 to 0.92. Following procedures used previously (NICHD Early Child Care Research Network 2003), we averaged sensitivity to non-distress, intrusiveness (reverse-coded), and positive regard to compute a sensitivity composite score at each wave. The Cronbach's alpha for sensitivity at 14 months was 0.70 and 0.89 at 24 months. The final maternal sensitivity score was the average of the two waves' scores, an approach consistent with existing longitudinal research of early maternal sensitivity and child attachment or other outcomes in the Western context (e.g., Dagan et al. 2021; Nivison et al. 2024; Steele et al. 2014; Waters et al. 2017).

2.2.2 | Attachment Script Assessment

The middle childhood version of the ASA is a narrative-based measure of secure base script knowledge among children (Waters, Bosmans, et al. 2015). By providing a list of prompt words indicating the beginning, middle, and end of a story, the assessment encourages children to describe fictional first-person accounts of seeking secure base support when distressed. If participants possess a well-developed secure base script, such knowledge should be activated by the prompt words, and the detail and elaborations in their story should match the temporal-causal structure of the secure base script (Waters and Waters 2021).

In this study, we created four new story outlines with the same narrative structure as the original middle childhood ASA (i.e., the child encounters a distressing situation during an activity with the parent; Waters, Bosmans, et al. 2015). In the English middle-childhood ASA, the outlines reflect scenarios more familiar to children in the West (e.g., *At the Beach* and *Scary Dog in the Yard*). The measure utilized in our current study features outlines more familiar to urban Chinese children (e.g., preparing for an examination, or getting lost in a crowded mall, see Appendix A). Consistent with the English and Dutch versions of the middle childhood

ASA, children were instructed to tell each story in the first-person (two stories with mom, two stories with dad), as if such an event happened between their mother or father and themselves. The entire procedure and all materials were presented in Mandarin. Each story was back-translated into English and confirmed with an experienced ASA researcher who was a native English speaker. Children were told that they did not have to use every word in the outline and did not have to follow the words in the order they were presented, but they were encouraged to tell the story with as many details as possible to make a good story. One outline about playing with a friend on a snow day was administered as the practice story and to limit hypothesis guessing.

Using the Waters, Bosmans, et al. (2015) seven-point secure base script scale for the middle childhood ASA, the ratings of the stories told by participants were based on how well they presented their secure base knowledge (see Appendix B). Scores of 7 were given to stories with rich secure base content (e.g., child signaling their distress, parents responding promptly and properly, the problem being resolved successfully, etc.). Scores of 3 were given to event-focused narratives with minimal/no secure base content. Scores of 1 were given to stories that contained atypical content inconsistent with a secure base script (e.g., parents causing distress or exacerbating the problem). Two independent coders (ICCs = 0.86–0.95) coded each story by storyline rather than individual participants so that coders were blind to the same participant's scores on other stories. All discrepancies were resolved through discussion. The Cronbach's alpha for these four stories was 0.77, and we used the average of the stories as the final score for ASA.

2.2.3 | Covariates

We included basic demographic variables as covariates in the regression analyses; this included parental reports of the child's biological sex and maternal education level collected at the age 6 months assessment. In the analysis, we coded sex as: 0 for male at birth and 1 for female at birth. For maternal education level, we coded this categorically: 1 = highest degree earned was a secondary school degree, 2 = highest degree earned was an undergraduate degree, and 3 = highest degree earned was a postgraduate degree.

3 | Analytic Strategy

3.1 | Missing Data

Given the mixed methods nature of the longitudinal study, at age 14 months, half of the participant pool was randomly selected and invited to participate in video-recorded observation tasks from which maternal sensitivity was measured, while the other half were invited to participate in a semi-structured interview. We conducted attrition analysis to compare whether families' demographic information varied significantly by child sex, maternal education at each wave, and data collection forms at each wave. Results indicated that families did not differ by any demographic factor (see Table S1). Furthermore, we used Little's Missing Completely at Random test to examine missingness in the dataset. Results showed that the data were missing completely at random and thus suitable for missing data procedures ($\chi^2(28) = 25.73$, $p = 0.587$). Based on the existing literature for optimal use of all available data points (Baker 2019; Graham 2009), we conducted all primary analyses with the full information maximum likelihood (FIML) algorithm to account for missingness, which is consistent with prior longitudinal studies that also used multiple waves of early maternal sensitivity to predict child outcomes in adolescence and adulthood (Fraley et al. 2013; Raby et al. 2015).

3.2 | Testing ASA Validity and the Association With Maternal Sensitivity

In the present study, we expected to observe a generalized secure base script that shaped variation across two relationship domains (mother–child and father–child). Following the approach of Waters, Fraley, et al. (2015), we tested this assumption using Confirmatory Factor Analysis. In the first step, we tested a generalized factor model that allowed the two latent factors (i.e., mother scripts and father scripts) to freely correlate, which aligns with the assumption that a higher-order factor (i.e., a generalized attachment representation) gives rise to covariation between the two domains. In step 2, we tested an alternative model in which secure base script knowledge was not generalized but rather was represented independently for each attachment relationship—fixing the correlation of the two domains to near zero (0.01). We then compared the model fits of the two models by conducting the chi-square difference test.

Following confirmation of the ASAs structure in the Chinese context, stepwise regression was used to examine the direct longitudinal association between maternal sensitivity and secure base script knowledge at age 10 years. Covariates were added to the regression model for robustness in the second step. The CFA and regressions were performed with the SEM command in STATA version 18 (StataCorp 2023). Missing data was accounted for with maximum likelihood with a missing value (mlmv option), which is equivalent to FIML (Enders and Bandalos 2001; Medeiros 2016).

4 | Results

4.1 | The Factor Structure of Child's Secure Base Script Knowledge

We conducted CFA to test the latent structure of ASA (bivariate correlations between items are presented in Appendix C—Table A1). Based on prior research, we hypothesized that the secure base script reflects a generalized knowledge structure, which is more so characteristic of the child rather than any specific attachment relationship (Waters, Bosmans, et al. 2015). When we allowed the two relationship domains to freely correlate, the data fit the model well (Hu and Bentler 1998): $\chi^2(1, N = 251) = 2.185$, $p = 0.139$, root-mean-square error of approximation (RMSEA) = 0.069, CFI = 0.995, TLI = 0.970. The estimated standardized correlation between the two factors was 0.94 (Figure 1). Then, we tested the alternative model where the correlation was fixed to 0.01 and the model fit was much poorer: $\chi^2(2, N = 251) = 108.854$, $p < 0.001$, RMSEA = 0.462, CFI = 0.551, TLI = −0.348. The chi-square test suggested that the fit of a generalized factor model was significantly better than

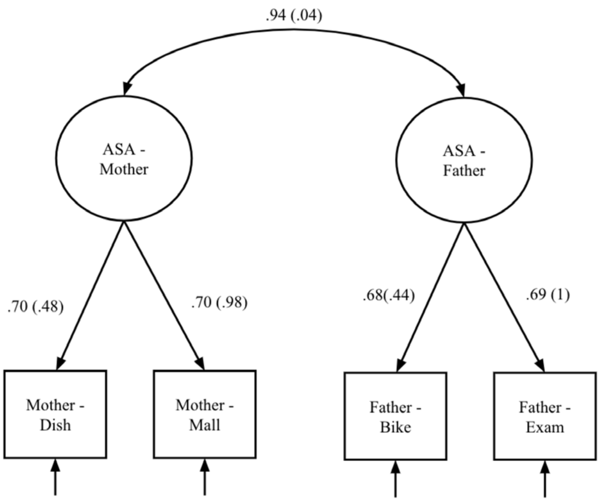

FIGURE 1 | Factor structure of Chinese middle childhood Attachment Script Assessment items. The general factor model allows for the generalization of secure base script knowledge across relationship domains (mother vs. father), the generalization path between the two domains was freely estimated (Waters, Bosmans, et al. 2015; Waters, Fraley, et al. 2015). The alternative model's path between the two domains was constrained to near zero. The model fit for the alternative model was significantly poorer than the general factor model. The estimated from the freely estimated model are shown outside the parentheses, and the estimates for the constrained model are presented inside the parentheses.

the alternative model, $\Delta\chi^2 = 106.67$, $p < 0.001$. This confirms that the mother and father storyline prompts in the Chinese middle childhood ASA tap into a generalized secure base script in a manner similar to that observed in Western contexts.

4.2 | Maternal Sensitivity and Child's Secure Base Script Knowledge

Bivariate correlations are presented in Table 1. At the mean level, ASA was positively correlated with maternal education and was also higher in girls than boys. Moreover, maternal sensitivity during infancy and toddlerhood was significantly associated with the child's overall secure base script knowledge at age 10 years ($r = 0.19$, $p = 0.006$). These associations are consistent with prior research based on Western samples (e.g., Steele et al. 2014; Waters et al. 2017).

To further analyze this association, we employed multiple regressions with FIML. The results of standardized coefficients and confidence intervals are presented in Table 2. In the first stage, we examined the basic model where maternal sensitivity was the only independent variable. Results suggested that the child's secure base script knowledge at age 10 years is positively associated with maternal sensitivity during infancy. In the second step, we entered child sex and maternal education level as covariates. The result suggests that accounting for child sex and maternal education level, maternal sensitivity at 14 and 24 months continued to significantly predict child ASA scores at age 10 years. In addition, results suggest that accounting for maternal level of education, child sex was also significantly associated with child ASA scores. Finally, given concerns related to

TABLE 1 | Correlations among Attachment Script Assessment Score (ASA), maternal sensitivity, mother's education, and child's sex with.

Variable	1	2	3
1. ASA (overall)	—		
2. Maternal sensitivity	0.19**	—	
3. Mother education	0.16	0.09	—
4. Child's sex	−0.20**	−0.07	−0.03
M	3.35	4.07	
SD	0.83	0.61	

Note: Child's sex: female = 0, male = 1. ASA = Attachment Script Assessment. Mother education: 1 = secondary school degree; 2 = undergraduate degree; 3 = postgraduate degree. Analyses conducted with pairwise deletion; ns ranged from 209 to 356.
Abbreviations: M, mean; SD, standard deviation.
*$p < 0.05$.
**$p < 0.01$.

TABLE 2 | Results of hierarchical multiple OLS regression models with FIML.

	β	p	95% CI	R^2
1. Maternal sensitivity	**0.20**	0.004	[0.06, 0.33]	0.04
2. Maternal sensitivity	**0.17**	0.012	[0.04, 0.30]	0.09
Child's sex	**−0.17**	0.005	[−0.29, −0.05]	
Mother education	**0.14**	0.023	[0.02, 0.26]	

Note: β is the standardized coefficient. Child sex: 0 = female, 1 = male. Mother education: 1 = secondary school degree; 2 = undergraduate degree; 3 = postgraduate degree. Statistically significant estimates are bolded. Analyses conducted with FIML; $N = 356$.

the theoretical and cultural relevance of the warmth construct (as well as a robustness check), we examined the predictive significance of the composite of the maternal sensitivity subscale alone in predicting ASA scores at age 10 years. Results did not substantively differ from those observed when the composite was used ($\beta = 0.20$, $p = 0.003$), suggesting that the inclusion of warmth and intrusiveness does not detract from the predictive significance of early care for later secure base script knowledge. To further probe associations between the three sensitivity subscales and secure base script knowledge, we tested whether those subscales differed significantly in the magnitude of their contributions to secure base script knowledge when examined as separate independent variables rather than being composited. Results indicated that there was no significant difference between the subscales and their association with later secure base script knowledge (see supplemental materials for further details). Finally, we examined if the sensitivity composites at 14 and 24 months differed significantly in the magnitude of their association with later secure base script knowledge in an effort to explore the potential impact of developmental timing on our observed effect. Again, results indicated no significant difference between sensitivity composites from different assessment waves and later script knowledge (see supplemental materials for further detail).

5 | Discussion

Attachment security is thought to arise from consistent and sensitive care from primary caregivers (i.e., sensitivity hypothesis). According to Bowlby (1969/1982), maternal sensitivity and the formation of attachment bonds are thought to be evolutionarily adaptive and thus a universal developmental process. However, Bowlby's assumptions about universality have long been questioned and criticized, with scholars arguing that maternal sensitivity is less applicable for child development in non-Western cultures (e.g., Keller 2018, 2019; Rothbaum et al. 2000). In our study, we sought to offer quantitative, long-term, longitudinal evidence for the cross-cultural validity of the sensitivity hypothesis, using data from a 9-year longitudinal study of child development conducted in Nanjing, China. China represented a distinctly non-Western caregiving context to evaluate the sensitivity hypothesis and its critiques, particularly along the warmth and autonomy-support axes. Notably, this study represents the first long-term longitudinal study of developmental antecedents of attachment representation outside of Western culture.

To evaluate attachment security at age 10 years, we constructed and validated a Chinese version of the middle-childhood ASA. As expected, all attachment-script items on the Chinese middle-childhood ASA (mother–child and father–child) converged on a single underlying factor. Consistent with prior research in Western contexts, our Chinese-adapted version of the ASA indicated that both mother and father items of the assessment effectively captured a fundamental construct of attachment security (see Waters, Bosmans, et al. 2015). After the preliminary validation of this measure, we tested a composite of maternal sensitivity from assessments at age 14 and 24 months as a predictor of attachment security (i.e., secure base script knowledge) at age 10 years. Our results supported the hypothesis that maternal sensitivity would significantly predict attachment security in Chinese children. Our data align with findings in prior research within the Chinese context that implicate knowledge of the secure base script as predictive of later psychosocial adjustment (i.e., peer conflict and loneliness; Yang et al. 2024). Furthermore, these results extend findings by previous Chinese research, which found a concurrent association between Chinese mothers' sensitivity and infant attachment security (Ding et al. 2012; Liang et al. 2021), providing the first long-term longitudinal evidence of the universality of the sensitivity hypothesis in China (Bowlby 1969/1982; Mesman et al. 2016; van Ijzendoorn and Sagi-Schwartz 2008). Furthermore, the observed effect size ($\beta = 0.20$) is consistent with prior research conducted in Western contexts as well as in cross-sectional studies in China (e.g., Nivison et al. 2024; Schoenmaker et al. 2015; Steele et al. 2014; Liang et al. 2021). Additionally, our correlational analysis revealed that children's attachment (ASA score) was positively correlated with maternal education. Given the highly verbal nature of the ASA task, our finding may be the result of more highly educated mothers engaging in more verbal stimulation with their children and story creation. This is in line with previous work indicating that maternal education is significantly associated with children's general story-telling/verbal ability (Bornstein and Haynes 1998; Fekonja-Peklaj et al. 2010). Girls also had higher ASA scores on average compared to boys. This finding is similarly in line with previous research showing a significant gender difference in ASA scores between adolescent girls and boys (Steele et al. 2014) and is consistent with the narrative literature more broadly (Bohanek and Fivush 2010).

The results from the present study suggest that the maternal sensitivity construct, originally developed by Ainsworth through her Uganda and Baltimore studies (Ainsworth 1967; Ainsworth et al. 1978/2015) and used widely thereafter in studies of attachment (Madigan et al. 2024), is also applicable to parent–child interactions in the Chinese context. Despite developmental literature characterizing Chinese parenting as less focused on expressing warmth and autonomy support (Lin and Fu 1990; Supple et al. 2009), our findings paint a different picture. Our data indicated that our Chinese sample scored relatively high on the composite of maternal sensitivity with a mean of 4.07 (SD = 0.61) on a scale ranging from 1 (lowest) to 5 (highest). From the three observed axes, mothers (on average) scored highly on positive regard and sensitivity to non-distress (M = 3.91, SD = 0.74; M = 3.82, SD = 0.77, respectively), and relatively low on intrusiveness (M = 1.62, SD = 0.65) across observations at ages 14 and 24 months. Our results suggest that maternal sensitivity, inclusive of warmth and autonomy support, is not only present in Chinese parenting but also appears to be a statistically significant variable in fostering secure attachment (i.e., secure base script knowledge) in Chinese children. We would argue that parental warmth, when appropriately aligned with the child's needs, reflects a facet of sensitivity and facilitates signaling and acceptance of support by the child. The increased frequency of these types of supportive interactions is then thought to be generalized into a cognitive script by the child via basic learning processes and carried forward across development (e.g., Waters and Waters 2024).

Contrary to the claims that maternal sensitivity is less relevant to children's development of attachment in cultures that diverge from Western conceptions of parenting and socialization goals (e.g., Keller 2018), our results suggest that *individual differences* in caregiving sensitivity in Chinese mothers predicted *individual differences* in child secure base script knowledge. However, research in a longitudinal Western sample indicated that secure base script knowledge also receives considerable input from other parental environmental variables, including reports of parental involvement and parental monitoring (Steele et al. 2014; Vaughn et al. 2016). Future studies should examine other culturally relevant parental and environmental factors that may explain additional variance in secure base script knowledge in Western and non-Western cultures alike.

Validation of the Chinese middle-childhood ASA presents an exciting opportunity to empirically study the impact of child attachment representations on psychosocial development within Chinese contexts, and potentially East Asia more broadly (e.g., Cheung et al. 2024). The ease of adaptation of the ASA presents the opportunity not only to further test the cross-cultural validity of the maternal sensitivity hypothesis but also to facilitate research on the nuanced relation between attachment representations and a spectrum of developmental correlates in varied cultural settings. Cultural critiques of maternal sensitivity extend beyond attachment security to include a wider array of developmental outcomes, such as child obedience and social competence (Keller et al. 2007). Research in Western populations gives us tentative insight into the links between maternal sensitivity, attachment security, and adaptive relationship functioning (e.g., peer or romantic relationships) as well as psychopathology (see Allen et al. 2007; Booth-LaForce and Kerns 2009; Groh et al. 2014; Mayseless and Scharf 2007 for review). Future studies could utilize the ASA to gain valuable insight into the relative significance of attachment representations in psychosocial developmental outcomes and relationship functioning across China and other non-Western cultures.

Our study contains several limitations that should be acknowledged. Given structural limitations in our data (i.e., only two waves of caregiving observations), we were unable to evaluate the impact of changes in maternal sensitivity over time or of sensitivity at different developmental stages on later secure base script knowledge. It still remains an open question whether attachment scripts are disproportionately influenced by caregiving during any particular phase of development or whether caregiving quality contributes equally across the infant, childhood, and adolescent periods. Some research suggests that secure base script knowledge is still open to revision through the adolescent period (Waters et al. 2022) but data on changes in sensitivity or timing related to caregiving quality and script development is limited (but see Eller et al. 2022).

Another limitation involves the assessment of maternal sensitivity. Sensitivity was measured in the context of free play interactions or semi-structured interaction tasks (e.g., book reading); thus, our coding of maternal sensitivity only captured responsiveness to children's cues in a non-distress context. Much of the literature on parental sensitivity and attachment relies on measuring sensitivity in non-distressing contexts (e.g., free play), but some have argued for the importance of distinguishing between sensitivity in distress versus non-distress contexts in favor of a more nuanced definition of sensitivity (e.g., Bernard et al. 2013; Goldberg et al. 1999; Leerkes et al. 2012). Although sensitivity to distress cues and non-distress cues tend to be at least moderately correlated (Leerkes et al. 2012), evidence suggests that sensitivity may differentially predict developmental outcomes depending on assessment approach, with sensitivity in the context of distress being especially predictive of secure attachment and later adjustment (Bornstein and Tamis-LeMonda 1997; Leerkes et al. 2012; Leerkes 2011; McElwain and Booth-LaForce 2006). The current study was limited by the use of previously collected observations that did not include interactions during distress-eliciting tasks; thus, future research should explore whether including sensitivity to distress, specifically, adds to the prediction of children's secure base script knowledge above and beyond sensitivity in a non-distress context. This may be especially relevant given that the ASA items focus specifically on secure base script knowledge in distress contexts.

The sample was representative of an upper-middle-class urban Chinese demographic. Yet, the Chinese parenting context also includes other populations with differing demographic characteristics. Previous research indicates that rural Chinese parents differ significantly from urban parents on the levels of positive encouragement, quality of caregiver–child relationships, and encouragement of initiative-taking (e.g., Chen and Li 2012; Han et al. 2023). Furthermore, Woodhouse et al. (2020) found that for families of low socioeconomic status (SES), maternal secure base provision was a significant predictor of infant attachment security whereas maternal sensitivity was not. Considering both the differences in caregiving practices and SES between rural and urban Chinese families, we suggest future research conducted in China consider the inclusion of rural samples. Moreover, our study focused on early interactions in the mother–child dyad given that our sample consisted mainly of families where biological mothers are the primary caregivers of the child. As a result, the generalizability of our results is limited for families where significant caregiving responsibilities may be shared with fathers, grandparents, nannies, or other relatives. In fact, recent research suggests that paternal caregiving behaviors play a salient and distinct role in child psychosocial development (Grossmann et al. 2002; Rice et al. 1997), including in Chinese contexts (Pan et al. 2020; Peng et al. 2022). Attachment relationships with non-parental caregivers (e.g., grandparents, Li et al. 2022; Liang et al. 2021) are also salient in Chinese children's socioemotional development. Future longitudinal projects should also study father and grandparent sensitivity to create a more complete picture of the caregiving network and to understand the contributions of each individual attachment relationship to the child's secure base script (Dagan and Sagi-Schwartz 2021). Another limitation of the present study is that we did not include an assessment of child verbal ability as it was not available in this dataset. We included child sex and maternal education in the analysis to control for variance in child language development, but future research would benefit from including an assessment of child language ability to account for potential confounds in measurement.

Taken together, our findings represent both a theoretical and methodological advance in our academic understanding of child social development in China. Our initial validation of the Chinese middle childhood ASA opens up exciting possibilities for understanding the antecedents and consequences of attachment development in the Chinese context. Our study also found empirical evidence to support the universality of the sensitivity hypothesis, an assumption that the attachment and cultural psychology communities have contended over for some time. Our findings represent only one study in the discourse surrounding cross-cultural understandings of maternal sensitivity and its consequences. More cross-cultural work needs to be conducted outside of the West in order to elucidate the universal applicability of maternal sensitivity and how it may interact with culture-specific components of development.

Disclosure

The analyses presented here were not pre-registered. The data necessary to reproduce the analyses presented here are not publicly available. The analytic code necessary to reproduce the analyses presented in this paper is not publicly accessible. The materials necessary to attempt to replicate the findings presented here are not publicly accessible. Deidentified data and analytic code can be made available upon request from the corresponding authors.

Data Availability Statement

The authors have nothing to report.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Appendix A

Chinese Middle-Childhood Attachment Script Assessment Prompts

Father Story 1 (Preparing for exam) 准备考试

在家 at home	我很担心 I worry	抬起头look up	睡觉的时间 bed time
准备考试 prepare for exam	爸爸 Dad	帮助 help	笑 laugh
难 difficult	读 read	说 say	睡 sleep

Father Story 2 (Riding a bike) 骑自行车

爸爸和我 Dad and I	骑 ride	爸爸 dad	抱 hug
天气很好 nice weather	摔下来 fall down	赶紧 hurry	家 home
自行车 bike	我受伤了 I got hurt	创可贴 band-aid	玩 play

Mother Story 1 (Going shopping) 逛商场

妈妈和我 mum and I	拥挤 crowd	妈妈找 mum find	抱 hug
商场 shopping mall	走散 lost	团圆 reunited	玩具 toy
逛 browsing	害怕 afraid	说 say	回家 home

Mother Story 2 (Preparing dinner) 准备晚餐

家 home	打碎 broke	哭 cry	亲 kiss
晚餐 dinner	手破了 cut hand	妈妈 mother	收拾 clean up
我端菜 I put dish on table	血 blood	创口贴Band-aid	吃饭 eat dinner

Appendix B

Examples of Attachment Script Assessment Scoring

Example 1 (High Scoring)

今天妈妈告诉我商场有活动, 我和妈妈便一起去逛商场。走到商场里, 看着那人山人海, 摩肩擦踵的人群, 我心里不禁有点害怕。走着走着, 我突然发现妈妈不在身边了。我心一急, 我的心里十分害怕, 都能感受到心在嘣嘣地跳。我四处寻找着妈妈。突然, 我看见一个很熟悉的身影, 再静静一看, 原来是妈妈。我以飞一样的速度跑过去, 跟妈妈抱在一起。妈妈说, "别害怕, 孩子, 没事的, 我们已经团圆了。"她抱着我, 买了玩具, 买了一些玩具安慰我, 就回家了。

Today mom told me there was an event in the mall, so my mom and I went to the mall together. When we walked in the mall, I saw so many people and was a bit afraid. We walked and walked, I suddenly found that my mom was not beside me. My heart felt anxious. I felt my heart was beating heavily. I looked for mom everywhere. Suddenly I saw a very familiar figure from the back. I quietly looked; it turned out to be mom. I ran to her in a way that seemed like I was flying and hugged her. Mom said, "Don't be afraid. My child, you are okay. We are together/united." She hugged me, bought some toys to assure/comfort me and then we went back home.

Example 2 (Low Scoring)

今天是周末, 学校不上课, 妈妈和我一起去逛商场, 然后我们走了很久, 也没有看到中意的东西, 这里面人很多, 十分的拥挤, 然后, 我和妈妈吵架了, 结果她往左边走了, 我以为她搭电梯了, 然后我就下去了, 然后我下电梯以后才发现跟妈妈走散了, 不知道在哪, 然后有一个保安就找到我, 说我妈妈正在找我, 然后她带着我回去, 妈妈说"以后你找不到我, 就在原地等, 不要乱跑"。然后我们一起回家了。

Today is the weekend, there is no school in session, mum and I went to shop in a mall, then we walked for so long, didn't see anything that interested me, there are many people here, it is very crowded. Then, mum and I got into a fight, as a result she turned to the left, I thought she went to take the elevator, so I went down, but after I went down in the elevator I realized that I was separated from mum, I didn't know where she was now. Then, a security guard found me, said mum was looking for me, then brought me back to her and mum said "in the future if you can't find me, just wait for me there, don't move around." Then we went home.

Appendix C

TABLE A1 | Pair-wise correlations across Attachment Script Assessment Score (ASA) storylines.

Variable	1	2	3	4
1. ASA (Mother—Dish)				
2. ASA (Mother—Mall)	0.48**			
3. ASA (Father—Bike)	0.42**	0.47**		
4. ASA (Father—Exam)	0.47**	0.42**	0.46**	
M	3.52	3.11	3.13	3.62
SD	1.13	0.98	1.02	1.20

Abbreviations: M, mean; SD, standard deviation.
**$p < 0.01$.